Tempus Known Problems and Solutions

Product Version 26.10
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Contents

About This Manual

This Known Problems and Solutions (KP&S) document describes important Cadence Change Requests (CCRs) for Cadence® Tempus™ Timing Solution and tells you how to solve or work around these problems.

1

Timing Analysis CCRs

Identified in Previous Releases

CCR 2509736: Huge mem jump during report_timing
CCR 2519983: GBA Reporting memory overhead

Description: In some cases, the Tempus software shows a memory increase in SOCV flow. This is due to a large number of `-max_paths` specified using the `report_timing` command. For example, the `report_timing -max_paths 1000000000 -nworst 1000` command will result in increased memory. This issue will be fixed in the next upcoming release.

Solution: For information only

CCR 1948868: Clock to clock co-relation shows up as "false" and "partial false" in DSTA mode

Description: The number of warning counts in DSTA mode (shown in the summary section of reports) may be greater than the number of warning counts in STA. This is expected behaviour.

If there are no warnings, then both STA and DSTA will not report any warnings.

Solution: You can generate a verbose report using the `check_timing` `-verbose` parameter. The report will show all the exact warnings and these match between STA and DSTA.

CCR 1422289: Performance degradation in do_extract_model

Description: Some fixes have been made to increase the accuracy of extracted timing models (ETM) by disabling parallel arc reduction. However, this fix may cause increase in runtime during ETM generation.

Solution: *For information only.*

CCR 588722: max_tran value is wrong on pins which have data phase and clock phase both

Description: The software might report incorrect max transition violations on mixed clock and data paths when you have specified the set_max_transition constraint with the -clock_path and -data_path parameters for a clock waveform. This is because when the data and clock signals on a pin are coming from two different launch points, the software considers max transition only for clock path and ignores any data path max transition when calculating the max transition violations.

Solution: *For information only.*

2

Base and SI Delay CCRs

Identified in Previous Releases

1549215: Messages to handle limitations with CMMMC

Description: Glitch analysis using CMMMC mode requires a specific view configuration.

For more details, see “Concurrent Multi-Mode Multi-Corner (CMMMC) Glitch Analysis” application note.

Solution: *For information only*.

CCR 1316234: create_spice_deck -run_path_simulation dumps empty column

Description: When `create_spice_deck -run_path_simulation` parameter is specified, the software may dump empty column for Spice delay/slew/arrival. This issue may occur in some cases when timing path begin point is register output pin - if create generated clock is asserted on a reg/Q pin. Such cases are not yet correctly handled by the newly supported feature, and will be fixed in a future release.

Solution: *For information only*.

CCR 1246277: report_noise -delay dump wrong attacker status (when -enable_delay_report & -enable_glitch_report turned ON together)

Description: `set_si_mode -enable_glitch_report true -enable_delay_report true` can give the incorrect attacker status in the `report_noise -delay` reports.

Solution: Run glitch analysis separate from Delay analysis when delay reports are desired (do not use both `enable_glitch_report true` and `enable_delay_report true` in the same run).

CCR 1096691: AAE giving incorrect Timing Window information when used report_noise -style extended

Description: Timing windows might differ from the ones shown using `report_delay_calculation -si`.

Solution: *For information only*.

CCR 1112087: report_double_clocking in AAE dump zero size report when used with EWM mode

Description: report_double_clocking in AAE is supported only in non-EWM mode.

Solution: *For information only*.

CCR 1042634: AAE SI : No SI analysis being performed in a testcase with read_sdf flow

Description: Flow of `read_sdf` cannot be used with AAE SI. AAE does not compute SI delay separately from base delay. When SDF file is provided, delay numbers are taken from SDF and no separate SI analysis is done.

Solution :*For information only*

CCR 934839: Nets with glitches exceeding 70% can have large differences in glitch calculation between AAE and CeltIC

Description: Nets with large glitches should be fixed prior to signoff. AAE delay may not match CeltIC delay for such cases.

Solution: *For information only.*

CCR 917323: Interconnect base delays are dumped as 0 with AAE turned ON while SGS reports it as 5ns

Description: This affects nets with more than 1000 receivers only.

Solution: *For information only.*

3

ECO CCRs

Identified in Previous Releases

CCR 2588979: Deserialize error while running optimization

Description: In previous release (21.10 or 21.11), the `eco_opt_design` command errors out when loading an ECO timing DB generated using 21.12 session.

Solution: Use 21.12 to run `eco_opt_design` with ECO timing DB generated from 21.12 session.

CCR 1621102: Mismatches seen in glitch fixing with xtalk/tran/cap fix when set to true and false.

Description: When fixing max_cap or max_tran, it may happen that a few new glitch violations are created after a regular buffer addition. Especially, during regular max_cap/max_tran fixing in ECO, the software cannot check for the creation of a new victim on a xcap net in its physical neighborhood. This is an area where code will be improved in a future release.

Solution: For information only.

CCR 1624719: Slew degradation with xtalk and DRV fixing

Description: In some cases, xtalk fixing may lead to slew degradation. This happens when there is a mismatch between incremental and full timing update.

Solution: Use `setDelayCalMode -combine_mmmc none` to resolve the difference.

4

General CCRs

Identified in This Release

CCMPR03149232 - KPNS – SSV 25.10 base release bin/checkSysConf patchData file update

Description:
In the SSV 25.10 Base Release, the patchData file for the checkSysConf is incorrect.

The patchData file is used to check whether a platform/OS is supported and what libraries are needed to run the binary.

The KPNS and CCR clarifies that the checkSysConf patch data file name for the tarkit at the download.cadence.com is provided incorrectly as “SSV241”. When the user runs the ckSysConf on RH7.4, it will return the OS as valid, which is incorrect as it is not a supported platform. For the SSV25.10 release, the supported platforms are RH8.4+, RH9.X, and SLES 15.

If the installation is updated based on the solution below to “SSV251” patchData file, the user will not see RH7.4 as a supported platform.

Solution:
Option 1:
If you do not want to install `confDataSSV25.10-p001.t.Z`, you can continue to run checkSysConf on the supported platforms RH8.4+, RH9.X, and SLES15.

Option 2:
If you need an updated patchData file, this can be picked from `/eng/tarkits/confDataSSV25.10-p001.t.Z` and uploaded it to the user site using the following commands:

```
cd <INSTALL_DIR>
rm -rf share/patchData
```
(Extract the tarkit).
```
tar zxvf confDataSSV25.10-p001.t.Z
```

For any further questions, contact Cadence Support.

Identified in Previous Releases

2692598: SSV 22.1 Tempus Copyright issue

Description: The current release shows Copyright year as 2021 when Tempus is launched. This will be fixed in the next 22.1x ISR release.

Solution: For information only.

1691166: Remove pt_to_celtic.tcl

Description: The timing windows file (TWF) containing "pt_to_celtic.tcl" is now replaced by "generate_twf.tcl". The "pt_to_celtic.tcl" file has been renamed to "generate_twf.tcl" as it describes its function better. There is no change in usage of the renamed file.

Solution: *For information only*.

CCR 1651415: Missing DMMMC support in write_def and save_design with merge_hierachical_def in Tempus

Description: The new `write_def` command, introduced in 16.20, is currently not supported in DMMMC mode. This feature may be supported in a future release.

Solution: *For information only*.

CCR 1348302: sdc sourcing multiple files using for each is failing in restore when used save restore portable mode

Description: When a user reads an sdc in which multiple constraint files are being sourced through a foreach loop , during `save_design -noTimingGraph` the sdc is saved with a variable with foreach substituted. This is because there is an existing variable with the same name defined in global name space. Therefore, restore design will fail.
For example:

```
sdc used in save session:
foreach tmp [list sdc1.new sdc2.new sdc3.new] {
source $tmp
}
```

```
sdc restored:
foreach tmp [list ${::IMEX::libVar}/mmmc/sdc1.new
${::IMEX::libVar}/mmmc/sdc2.new ${::IMEX::libVar}/mmmc/sdc3.new] {
source ${::IMEX::libVar}/mmmc/sdc3.new (<<< issue as only 3rd file will be sourced all three
times)
}
```

Solution: To avoid this, update sdc script to use following instead.

```
source sdc1.new
source sdc2.new
source sdc3.new
```

CCR 903576: Change in rail analysis results for VDD/VSS due to extraction.

Description: In this release, there is a difference in the rail analysis results for VDD and VSS nets. The difference in results between the previous and current versions of the software is due to difference in propagation. This issue will only affect designs using activity propagation that have empty modules.

Solution: Previously, the empty module was handled as a hierarchical module by default, and now the empty module is handled as a dummy cell/leaf instance. To get the original results, ensure that you handle the empty module as a hierarchical module by using set_import_mode -keepEmptyModule true. Note that the new results (without setting -keepEmptyModule true) are better as the pin directions on the empty module are now correct and match the Verilog.